Explainable Spatiotemporal AQI Prediction

Prior-Work Review and Novelty Analysis

Working paper — internal positioning document

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1. About this document

This document reviews prior work relevant to the proposed research project on explainable spatiotemporal Air Quality Index (AQI) prediction, and assesses which parts of the proposed contribution are genuinely novel, which are already established, and how the project should be positioned to survive peer review. It is a positioning document, not the paper itself.

How it was produced

- Structured web literature search (English-language, US region) conducted September 2026 across the attention-as-explanation debate, explainability in air-quality modelling, spatiotemporal multimodal AQI/PM2.5 prediction, faithfulness-evaluation methodology, and concept-based explanation.
- Sources were triaged from search results and abstracts. Several entries were read only at the abstract/snippet level, not full text.

Limitations — read before citing anything here

- **Not a systematic review.** No PRISMA protocol, no exhaustive database sweep (Scopus / Web of Science / IEEE Xplore were not queried directly). Treat coverage as indicative, not complete. Section 12 gives concrete queries to run yourself.
- **Verify every reference before use.** Some titles returned by search carried 2025–2026 publication years that should be confirmed against the publisher of record; a minority may be preprints, in-press items, or index artefacts. Confirm authors, venue, year and DOI from the primary source.
- **Absence of evidence is not evidence of absence.** Where this document says a combination "appears under-explored", that means the search did not surface it — not that it does not exist. Novelty claims in the paper must be backed by your own systematic search.
- **The design is multi-city; the data on disk is single-city.** The intended study spans multiple cities across climatic and emission regimes, with cross-region (leave-one-city-out) generalisation built into the research questions. Files on disk currently cover one region only: satellite AOD, HCHO and NO_2, ERA5 meteorology (2 m temperature, 2 m dewpoint, surface pressure, 10 m wind u/v, total precipitation), and lat/lon, at daily resolution for 2023, plus CPCB station AQI (Alipur, Delhi). CO, SO_2, O_3, boundary-layer height, elevation and land use are also absent. Assembling aligned multi-city data is a prerequisite for the stated contribution and is the first major task (see Appendix A).

2. Executive summary — novelty verdict

The project's framing question — **does attention faithfully represent what actually drives the model's AQI predictions?** — is the right question, and it is not the standard framing in the air-quality literature. However, several individual ingredients are well established and cannot be presented as new.

Established (do not claim as novel)

- Multimodal spatiotemporal deep learning for AQI/$PM_{2.5}$ that fuses satellite retrievals, meteorology and ground observations — a large and active literature, including near-identical architectures (LSTM + attention + satellite + meteorology + historical AQI).
- Attention layers inside air-quality forecasting models — common.
- Applying SHAP / LIME / partial dependence to air-quality models — common, though mostly on tree ensembles and CNN-LSTM.
- Comparing an attention layer's feature ranking with a post-hoc SHAP ranking on an air-pollution model, and observing partial disagreement — already done at least once (NO_2 forecasting).
- The general result that attention weights are not, in themselves, faithful explanations — established since 2019 in NLP and echoed in graph / tabular settings. The paper cannot claim to discover this.
- Faithfulness metrics (comprehensiveness / sufficiency, remove-and-retrain, deletion curves), the XAI disagreement problem, TCAV, and the meteorological air-stagnation index — all established.

Open / under-explored (candidate contributions)

1. A **systematic faithfulness audit of attention carried into the multimodal, spatiotemporal, regression setting** for AQI. The attention debate is overwhelmingly NLP, classification, and single-modality; continuous AQI regression with cross-modal and cross-time attention is a materially different regime where faithfulness has not been carefully characterised.
2. **Multi-method triangulation adjudicated by physical plausibility.** Comparing attention against SHAP, LIME, permutation/perturbation importance and concept testing, and then resolving disagreements against known atmospheric relationships (dispersion vs wind speed, accumulation under stagnation, boundary-layer venting, hygroscopic growth with humidity) used as an external reference. Most domain papers use one or two methods and treat agreement as validation.
3. **Regime- and region-conditioned faithfulness.** Testing whether attention's faithfulness and stability degrade at high-AQI extremes, across seasons (e.g. plains winter stagnation, biomass-burning episodes), across cities, and across climatic zones — including whether explanations stay faithful for a city held out of training. Tail/shift behaviour of explanations is understudied generally and specifically for AQI.
4. **Concept-level faithfulness with a physically grounded concept.** Using concept activation testing with “atmospheric stagnation” defined from air-stagnation-index components to test whether the model's latent representation is sensitive to stagnation during high-AQI predictions, and whether that agrees with feature-level attention and SHAP. Concept-based XAI is rare in air-quality ML.

5. **Ablation × explanation co-analysis.** Linking the data-source / architecture ablation ladder to whether explanations become more physically coherent as components are added — i.e. does a modality that improves RMSE also improve explanation faithfulness?

One-line positioning

"Not another explainable-AQI model, but an audit of whether the explanation you get for free from attention is telling you the truth — tested against independent XAI methods and against atmospheric physics, across pollution regimes."

3. Strand A — Spatiotemporal AQI prediction from satellite and meteorological data

This strand is crowded and fast-moving. It establishes that the predictor architecture proposed here is not, by itself, a contribution.

3.1 Multi-source fusion and satellite-to-surface estimation

- Machine-learning estimation of surface pollutant concentrations from satellite column retrievals plus reanalysis meteorology is a mature sub-field. Surface NO_2 over Europe has been estimated from Sentinel-5P TROPOMI plus meteorology with gradient-boosted trees at ~1 km resolution (Remote Sensing of Environment).
- Region-specific work already exists, including for Indian cities: Sentinel-5P + ML for spatiotemporal NO_2 in Delhi (including the COVID-19 lockdown signal); Sentinel-5P-driven ozone sub-index and composite AQI estimation explicitly under the CPCB National Air Quality Index framework; and improved daily $PM_{2.5}$ reconstructions across India. Most such work is single-city or single-country and does not evaluate cross-city transfer.
- Implication: satellite + reanalysis + station fusion for regional AQI is an existing baseline, not a novelty. The national-AQI sub-index computation is already in the literature — cite it rather than presenting it as your own construction. Multi-city coverage with an explicit leave-one-city-out transfer evaluation is the less-trodden part.

3.2 Model families

- **Recurrent:** LSTM / GRU and CNN-LSTM hybrids remain the workhorse for pollutant time series.
- **Transformer / efficient-attention:** SpatioTemporal-Informer and related long-sequence Transformers have been applied to $PM_{2.5}$ with spatiotemporal structure.
- **Graph-based:** Graph Attention Networks, often coupled with GRUs, model inter-station spatial dependence for $PM_{2.5}$ and composite pollution; multi-source spatiotemporal graph-attention networks incorporate satellite and meteorological inputs.
- **Interpretable-by-design forecasters:** the Temporal Fusion Transformer combines variable-selection networks with interpretable multi-head attention and is frequently promoted as an interpretable multi-horizon model — directly relevant as both a baseline and a foil for the faithfulness question.

3.3 Closest prior work on the predictor

A Novel Approach for Predicting the Air Quality Index of Megacities through Attention-Enhanced Deep Multitask Spatiotemporal Learning (arXiv:2407.11283, 2024) is the nearest neighbour: an LSTM-based, attention-enhanced, multitask spatiotemporal model consuming satellite imagery, meteorological data and historical AQI to forecast AQI and individual pollutants, with model-interpretation discussion aimed at policymakers.

- **Risk:** a reviewer will ask how the proposed predictor differs. It largely does not need to — but the paper must state plainly that the predictor is a re-implementation / adaptation of an established

design, and that the contribution is the explainability audit. Consider using this architecture as one of the compared attention models, or reproducing it as a reference point.

3.4 Takeaway for Strand A

Frame the model section as engineering scaffolding for the research question, benchmarked honestly against baselines (Linear, RF, SVR, XGBoost, MLP, LSTM/GRU, Transformer/TFT, attention vs no-attention). Do not claim architectural novelty; claim a controlled testbed.

4. Strand B — Explainability in air-quality prediction

This strand shows that "explainable AQI" is itself crowded, and that the wedge must be faithfulness, not explainability.

- SHAP is now routine for AQI/pollutant models: ensemble / gradient-boosting / XGBoost / stacking pipelines with SHAP-based global and local attribution; hybrid Random-Forest + ARIMA models with SHAP; CNN-LSTM AQI forecasting reported with combined SHAP, LIME and partial-dependence visual analysis; interpretable-ML frameworks for urban air quality.
- Reviews of ML for air-quality prediction now treat interpretability as a standard reporting expectation and note that black-box structure obscures pollutant-formation and transport mechanisms.
- **Ante-hoc + post-hoc on the same model already exists.** Work on NO_2 forecasting combines an attention layer (ante-hoc) with post-hoc SHAP: the attention layer emphasised CO and NO_2, while SHAP ranked NO_2, then $PM_{2.5}$, then CO. That is a small, direct instance of "attention and SHAP disagree on an air-pollution model." The proposed project must cite this and clearly exceed it: multiple XAI methods rather than two; formal faithfulness metrics rather than ranking eyeballing; regime conditioning; concept-level analysis; and physical adjudication of disagreements.
- Spatiotemporal "feature-interpretable" air-quality models exist that expose per-feature, per-time contributions — again, positioned as interpretable, rarely audited for faithfulness.

4.1 Takeaway for Strand B

The gap is not "add SHAP to an AQI model" or "add attention to an AQI model." It is: given that domain papers routinely present attention and/or SHAP as explanations, test whether those explanations are faithful and physically meaningful, and characterise when they fail.

5. Strand C — The attention-as-explanation debate

This is the intellectual core the project draws on. The literature is mature but almost entirely outside the air-quality / regression / multimodal setting.

Work	Position / contribution
Jain & Wallace (2019), "Attention is not Explanation", NAACL	Attention weights often correlate poorly with gradient-based importance, and alternative attention distributions can yield near-identical predictions — so attention is not a faithful explanation in the classifiers tested.
Wiegreffe & Pinter (2019), "Attention is not not Explanation", EMNLP	Whether attention explains depends on the definition and on testing the whole model; proposes uniform-weight and adversarial-training diagnostics. Attention can be plausible and sometimes usable.
Serrano & Smith (2019), "Is Attention Interpretable?", ACL	Erasing / permuting attention weights barely moves predictions; gradient-ranked attention predicts impact better than raw magnitude. Attention is a noisy, not fail-safe, importance signal.
Bastings & Filippova (2020), BlackboxNLP	If the goal is input attribution, saliency methods are the better-motivated tool than attention.
Abnar & Zuidema (2020), "Quantifying Attention Flow", ACL	In stacked layers, token information mixes, so raw attention is an unreliable probe; attention rollout / flow correlate better with ablation and gradient importance.
Bibal et al. (2022), "Is Attention Explanation?", ACL	Survey of the debate; formalises the faithfulness vs plausibility split and the conditions under which attention has been argued to explain.
Liu et al. (2022), "Faithfulness Violation Test", ICML	Proposes a test across attention-based explanation methods; finds frequent violations of the assumptions needed for attention-derived explanations to be faithful.

5.1 What this means for the project

- The **faithfulness vs plausibility** distinction must be stated explicitly and used throughout: an explanation is plausible if it looks reasonable to a human; faithful if it reflects the computation that produced the output. The project targets faithfulness.
- **Novelty caveat:** "attention is not automatically explanation" is a settled prior, not a finding. The contribution is *characterising where attention is and is not faithful in this specific, applied, multimodal spatiotemporal regression setting*, and whether independent methods and physics agree.
- **Method to adopt:** don't compare raw attention only. Include attention rollout / flow (or gradient-weighted attention) as Abnar & Zuidema recommend, so a negative result cannot be dismissed as "you used naive attention."
- **Gap to assert (with your own search backing):** the search surfaced no rigorous transfer of this debate to satellite+meteorological spatiotemporal AQI regression with cross-modal attention. NLP classification findings do not automatically transfer — different architecture, continuous target, correlated physical inputs, temporal lags.

6. Strand D — Faithfulness evaluation methodology

The project needs defensible metrics. These are the standard tools; adopt and cite them rather than inventing ad-hoc measures.

6.1 Perturbation / ablation family

- **Comprehensiveness and sufficiency (ERASER; DeYoung et al., 2020).** Comprehensiveness = drop in model output when the top-k "important" features are removed; sufficiency = how well the top-k alone reproduce the output. Aggregated over k as area-over-the-perturbation-curve (AOPC). Directly portable to AQI regression by substituting Δprediction for Δclass-confidence.
- **Deletion / insertion curves.** Progressively remove (or add) features in importance order and track error; a faithful ranking degrades prediction fastest when its top features are deleted.
- **Remove-and-Retrain — ROAR (Hooker et al., 2019).** Remove the top features, retrain, measure accuracy loss. Controls for the out-of-distribution artefact that plain perturbation suffers from. Expensive but the gold-standard control; several popular attributions underperformed random under ROAR.
- **Design note for correlated inputs:** wind u/v, temperature, dewpoint, pressure and AOD/NO_2 are physically correlated. Single-feature perturbation leaks information through correlates. Use grouped / conditional perturbation, and consider knockoff-style or model-retraining controls; report this explicitly.

6.2 Time-series-specific attribution and evaluation

- **Dynamic Masks (Crabbé & van der Schaar, 2021)** — learn a perturbation mask over time steps/features with a faithfulness objective; a strong baseline attributor and evaluation lens for temporal inputs.
- **FIT (Tonekaboni et al., 2020)** — instance-wise feature importance over time via distributional shift; designed for the "what mattered and when" question the project asks of historical inputs.
- Time-frequency attribution work indicates faithfulness scores can differ between time-domain and frequency-domain perturbations — worth noting given diurnal/seasonal AQI cycles.

6.3 Cross-method disagreement

- **The Disagreement Problem (Krishna et al., 2022).** LIME, KernelSHAP and gradient methods routinely disagree on top features and their order; practitioners lack principled tie-breakers. Provides a formal metric suite (feature agreement, rank agreement, sign agreement, signed-rank agreement) — use these to quantify attention-vs-XAI overlap rather than informal comparison.
- Consequence for the project: disagreement between attention and SHAP is *expected*; the research value is in explaining it (Q6) and adjudicating it against physics, not merely reporting it.

7. Strand E — Concept-based explanation and physically grounded concepts

7.1 TCAV

- **TCAV (Kim et al., 2018)** learns a concept activation vector as the normal to a linear boundary separating concept-present from concept-absent examples in a hidden layer, then uses directional derivatives to measure how much that concept increases the likelihood of a prediction. Extensions address cross-layer consistency (global CAV frameworks) and non-image modalities (e.g. concept-based explanation for an EEG Transformer).
- The search surfaced essentially no application of TCAV to air-quality or atmospheric-composition prediction. This is a genuine opening — but also a warning that methodological care is needed (concept dataset construction, statistical testing, layer choice, confounding between concept and label).

7.2 Atmospheric stagnation as a ready-made physical concept

- The **Air Stagnation Index** is an established meteorological construct: stagnation is diagnosed from low 10 m wind speed, low mid-tropospheric wind, and little precipitation; ventilation-index variants use the product of boundary-layer height and near-surface wind.
- Stagnation is causally linked to particulate accumulation, including quantified impacts on Indian surface $PM_{2.5}$ and Chinese haze episodes.
- This gives the project a concept with (a) a physical definition, (b) known directional effect on AQI, and (c) constructible concept/anti-concept sets from ERA5 fields available for any city (wind u/v, precipitation, boundary-layer height). Testing whether the model's representation is sensitive to“stagnation” during high-AQI predictions — and whether that sensitivity is consistent across cities and tracks feature-level attention and SHAP on wind/PBL — is a concrete, defensible extension.
- **Caveat:** the current dataset lacks boundary-layer height; either add ERA5 `boundary_layer_height`, or define the concept from the available wind + precipitation + humidity proxy and acknowledge the reduced fidelity.

8. Synthesis — coverage map

Each row is a dimension of the proposed work. “Coverage” summarises what the search found; “Headroom” is where this project can contribute.

Dimension	Prior coverage found	Headroom for this project
Multimodal spatiotemporal AQI predictor (satellite + met + spatial + history)	Extensive; near-identical LSTM+attention multitask model exists (arXiv:2407.11283); Transformer/Informer, GAT-GRU variants	Low — use as testbed, not contribution
Attention inside AQI models	Common	Low
SHAP/LIME/PDP on AQI models	Common (mostly trees, CNN-LSTM)	Low on its own
Attention vs SHAP compared on an air-pollution model	Exists in small form (NO_2 forecasting, 2-method ranking comparison)	Medium — extend to many methods + formal metrics
Attention-as-explanation faithfulness, rigorously tested	Mature in NLP classification; echoed in GNN/tabular	High in the AQI / regression / multimodal setting
Faithfulness metrics (compreh./suff., ROAR, deletion) applied to AQI attention	Rare in this domain	High
Regime-conditioned faithfulness (extremes, seasons)	Not found	High
Cross-city / cross-regime consistency & transfer of attention faithfulness (incl. held-out city)	Not found	High
Physical-plausibility adjudication of XAI disagreements	Not found as a systematic criterion	High
TCAV / concept testing in air-quality ML	Not found	High (with methodological care)
“Atmospheric stagnation” as a TCAV concept	Not found; index itself well established	High
Ablation ladder × explanation-quality co-analysis	Not found	Medium–High

9. Novelty assessment

9.1 Defensible novelty claims

1. **First systematic faithfulness audit of attention in a multimodal, spatiotemporal AQI regression model**, using established faithfulness metrics rather than ranking comparison, and covering cross-feature and cross-time attention. (Condition on your own systematic search confirming no direct precedent.)
2. **A physical-plausibility criterion for adjudicating XAI disagreement.** Where attention, SHAP, LIME and perturbation disagree, the tie-break is consistency with known atmospheric relationships — turning the disagreement problem from a nuisance into an evaluable question.
3. **Regime-conditioned explanation analysis.** Quantifying whether attention faithfulness and stability degrade at high-AQI extremes and across seasons — i.e. explanation behaviour under the distribution shift that matters most operationally.
4. **Cross-city consistency and transfer of explanations.** Measuring whether attention rankings and their faithfulness hold across cities and climatic regimes, and whether explanations for a city held out of training remain faithful and physically plausible — to our knowledge not previously quantified for AQI.
5. **Concept-level faithfulness for a physically grounded concept (atmospheric stagnation)** via concept activation testing, cross-checked against feature-level methods and across regimes.
6. **Explanation quality as an axis of the ablation study**, not just RMSE/R^2 — testing whether components that improve accuracy also improve faithfulness/physical coherence.

9.2 Claims to avoid or soften

- ✗ "We propose a novel attention-based multimodal spatiotemporal AQI model." — architecture is not novel.
- ✗ "We show attention is not a faithful explanation." — already established; you are testing *whether and when* it holds here.
- ✗ "First to use SHAP/LIME for AQI explainability." — far from first.
- ✗ "First to combine ante-hoc attention with post-hoc SHAP for air pollution." — a small prior instance exists (NO_2).
- ⚠ "Attention provides explainability." — the project's own thesis forbids this phrasing; keep it as a hypothesis under test.

9.3 Principal novelty risks

- **Architectural proximity to arXiv:2407.11283.** Mitigation: explicitly cite it, describe your predictor as an adaptation, and locate the contribution in the audit. Optionally reproduce it as a compared model.

- **“Yet another explainable-AQI paper” reception.** Mitigation: lead with the faithfulness question and the physics-adjudication design in the title and abstract; keep SHAP/LIME framed as instruments, not contributions.
- **Data on disk is single-city; the design is multi-city.** The stated contribution (cross-city consistency, leave-one-city-out transfer) requires assembling aligned data for several cities across regimes — currently only one region is present. Mitigation: treat multi-city data assembly as work-package one; pick 6–12 cities spanning climatic zones; add boundary-layer height for the stagnation concept. Optionally still add $CO/SO_2/O_3$ if pollutant-level attention is wanted.
- **Negative-result framing.** If attention turns out mostly faithful, or mostly not, either way is publishable *only if* the evaluation is rigorous and the conditions are characterised. Pre-register the metrics and thresholds.

10. Recommended positioning and contribution statement

10.1 Draft contribution paragraph

“We do not assume that attention weights are explanations. We treat the correspondence between attention and predictive importance as an empirical hypothesis and test it in a multimodal spatiotemporal AQI regression model that fuses satellite retrievals, reanalysis meteorology, location and historical AQI. Attention (raw and flow-corrected) is compared against SHAP, LIME and grouped permutation/perturbation importance using established faithfulness metrics — comprehensiveness, sufficiency, deletion curves and remove-and-retrain — and against concept activation testing for the physically defined concept of atmospheric stagnation. The model is trained and evaluated across multiple cities spanning distinct climatic and emission regimes, both within-city and under a leave-one-city-out protocol. We characterise where attention is and is not faithful across pollution regimes, seasons, cities and climatic zones, measure how faithfulness transfers to cities unseen in training, quantify agreement and disagreement among methods with formal metrics, and adjudicate disagreements against known atmospheric relationships. We further fold explanation faithfulness into the ablation study, testing whether data sources and architectural components that improve accuracy also improve explanation quality.”

10.2 Suggested title directions

- “Is Attention Telling the Truth About Air Quality? A Faithfulness Audit of a Multimodal Spatiotemporal AQI Model”
- “Faithful or Merely Plausible? Comparing Attention with SHAP, Perturbation and Concept Tests for Explainable AQI Prediction”
- “When Attention and Physics Disagree: Regime-Conditioned Explanation Faithfulness in Satellite-Meteorological AQI Modelling”
- “Does It Travel? Cross-City Consistency of Attention-Based Explanations in Multimodal AQI Prediction”

10.3 Candidate venues

- **Domain:** Atmospheric Environment; Science of the Total Environment; Environmental Modelling & Software; Air Quality, Atmosphere & Health; Environmental Pollution; IEEE TGRS; Remote Sensing of Environment.
- **XAI / ML:** workshops and tracks on interpretability (e.g. XAI-focused venues, FAccT), or ML-for-environment tracks at NeurIPS/ICML/KDD.

11. Gaps this paper can claim to fill

Phrase each as an explicit gap statement in the introduction, each backed by your own systematic search.

Gap statement	Supporting rationale from this review	Paper element that fills it
The attention-faithfulness debate has not been carried rigorously into multimodal spatiotemporal AQI regression.	Debate literature is NLP/classification; AQI-attention papers do not test faithfulness.	RQ1–RQ2; faithfulness-metric experiments on attention vs XAI.
Domain XAI-for-AQI studies rarely quantify cross-method (dis)agreement or resolve it.	SHAP/LIME used as validation; disagreement problem not engaged.	Formal agreement metrics (Krishna et al.); disagreement case studies (RQ5–RQ6).
Explanation behaviour under pollution extremes and seasonal shift is uncharacterised for AQI.	No regime-conditioned faithfulness work found.	RQ3–RQ4; stratified evaluation by AQI band, season, station.
Whether attention explanations are consistent across cities/climates, or transfer to an unseen city, is untested for AQI.	Domain XAI work is single-city; no cross-region faithfulness evaluation found.	RQ7; leave-one-city-out evaluation of faithfulness and physical plausibility.
No external, physically grounded reference is used to judge AQI explanations.	Plausibility judged informally, if at all.	Physical-relationship checklist; stagnation concept via TCAV.
Concept-based explanation is essentially absent from air-quality ML.	No TCAV-in-air-quality work found.	Concept activation testing for atmospheric stagnation.
Ablation studies optimise accuracy only, ignoring whether added components explain better.	Ablation ladders report RMSE/R^2 only.	Ablation × faithfulness co-analysis.

12. Recommended further literature search

Run these before finalising any novelty claim. Use Scopus, Web of Science, IEEE Xplore, ACL Anthology, Semantic Scholar and arXiv; also inspect the forward citations of Jain & Wallace (2019) and Bibal et al. (2022) filtered to environmental / remote-sensing / time-series terms.

Query strings

- (“attention” AND (“faithful*” OR “explanation” OR “interpretab*”) AND (“air quality” OR “PM2.5” OR “PM10” OR “AQI” OR “air pollution” OR “ozone” OR “NO2”))
- (“attention is not explanation” OR “attention as explanation”) AND (“time series” OR “regression” OR “forecasting” OR “remote sensing”)
- (“SHAP” OR “LIME” OR “Shapley”) AND “attention” AND (“air quality” OR “PM2.5” OR “AQI”) AND (“compar*” OR “agreement” OR “faithful*”)
- (“TCAV” OR “concept activation vector” OR “concept-based explanation”) AND (“atmospher*” OR “air quality” OR “air pollution” OR “climate” OR “meteorolog*” OR “weather”)
- (“air stagnation index” OR “ventilation index” OR “stagnation”) AND (“machine learning” OR “deep learning” OR “neural network” OR “explainab*”)
- (“multimodal” OR “multi-source” OR “data fusion”) AND “spatiotemporal” AND (“AQI” OR “air quality” OR “PM2.5”) AND (“attention” OR “transformer”) AND (“interpretab*” OR “explainab*”)
- (“faithfulness” OR “comprehensiveness” OR “sufficiency” OR “ROAR” OR “remove and retrain”) AND (“time series” OR “environmental” OR “air quality”)
- (“Sentinel-5P” OR “TROPOMI”) AND (“AQI” OR “National Air Quality Index” OR “CPCB”) AND (“India” OR “Delhi”) AND (“explainab*” OR “interpretab*” OR “attention”)
- (“cross-city” OR “multi-city” OR “multi-region” OR “leave-one-city-out” OR “spatial transfer” OR “geographic generali*ation” OR “domain generalization”) AND (“air quality” OR “PM2.5” OR “AQI”) AND (“deep learning” OR “attention” OR “neural”)
- (“explanation” OR “feature importance” OR “SHAP” OR “attention”) AND (“stability” OR “consistency” OR “transfer” OR “robustness”) AND (“across cities” OR “across regions” OR “out-of-distribution” OR “unseen”) AND (“air” OR “environmental” OR “time series”)

13. References (grouped; verify before citing)

URLs are the search landing pages used during this review; replace with the DOI / publisher of record when citing. Publication years marked "[verify]" were reported by search with a recent date that should be confirmed.

Attention-as-explanation debate

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Faithfulness-evaluation methodology

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Appendix A — Data on disk vs the multi-city design

Snapshot of what is on disk versus what the multi-city design needs. Assembling the aligned multi-city dataset is work-package one.

Category	Design needs	Present on disk now	Action
Geographic coverage	6-12 cities spanning climatic / emission regimes (plains stagnation, coastal ventilation, semi-arid dust), for leave-one-city-out transfer	One region only (Delhi grid)	Select the city set; run the retrieval + alignment pipeline for every city; record each city's climatic zone.
Satellite composition	NO_2, CO, SO_2, O_3, and others	AOD, HCHO, NO_2 (daily, 2023, gridded ~Delhi)	Add Sentinel-5P CO, SO_2, O_3, (HCHO already present) if pollutant-level analysis is wanted; else scope to NO_2/AOD/HCHO.
Meteorology	temp, humidity, wind, pressure, precip, boundary-layer height	ERA5: 2 m temp, 2 m dewpoint, surface pressure, 10 m wind u/v, total precipitation	Add ERA5 boundary_layer_height (needed for the stagnation concept); derive RH from temp+dewpoint; pull for every city.
Spatial	lat, lon; optionally elevation, land use	lat, lon	Add a DEM (e.g. SRTM) and a land-use/land-cover layer only if spatial-context claims are made.
Ground truth	many stations across many cities	CPCB hourly AQI, Alipur (Delhi); some 2017 data	Pull agency stations for every city in the set; keep station id for within-city stratification.
Temporal resolution / span	consistent daily series over a common multi-year window across all cities	Daily satellite/met; hourly station AQI; 2023 (plus stray 2017)	Fix the shared date window; aggregate station AQI to daily; document gap/mismatch handling.

Consequence for the paper: the current data is a single region (Delhi), 2023, daily resolution, on NO_2 / AOD / HCHO + ERA5 meteorology. The design (cross-city consistency, leave-one-city-out transfer, regime conditioning) cannot be run on single-region data, so the first deliverable is an aligned multi-city dataset over a common window; the stagnation concept also needs the ERA5 boundary-layer-height field. CO/SO_2/O_3, land use, elevation and multi-year extension remain optional / future work.